Soft-device closure for the h-model of actin–myosin interaction: notation, imposed quantities, protocols and results

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Abstract

We implement the h-model of Chaintron, Caruel and Kimmig (Math. in Action 12, 2023) with N heads and add a closure for the sliding variable x_c in which the total force is imposed and the sliding speed is an output (soft device). Section 2 defines every symbol, and marks those that are not in the paper. The note then states what is imposed and what is measured in each numerical experiment, defines every plotted quantity, and reports the results: the force–speed relation with its S-shaped part when lengthening is allowed, and stick–slip cycles in a mixed device (a spring in series, far end driven). Calibration choices that are inferred rather than read from the paper are listed in Section 9.

1 Setting

The paper computes the force–velocity curve in *velocity clamp*: the sliding velocity $\dot{x}_c$ is imposed and the mean force per head is measured (its Fig. 5.2). In a force experiment the roles are reversed. Here we do both, with the same head-level model:

- velocity clamp gives the steady-state force $\tau_{ss}(u)$ at imposed shortening speed $u = -\dot{x}_c$;
- the soft device imposes a load, adds a macroscopic viscosity ν on x_c, and measures $u(t)$;
- the mixed device adds a spring between x_c and a far end that is held or driven.

The first two are linked by a steady-state relation (Section 5) whose non-monotone part produces the S.

2 Notation and conventions

Sign convention. Following the paper, $\dot{x}_c < 0$ is contraction. All forces are counted positive in the direction of contraction (toward $-x$), unless stated. A force along $+x$ is therefore the opposite of what is called "positive" here. Units are nm, ms, pN, zJ, with $1\,\mathrm{pN\,nm} = 1\,\mathrm{zJ}$ and $1\,\mathrm{nm/ms} = 1\,\mu\mathrm{m/s}$.

Origin column. "P" means the symbol is in the paper (Math. in Action 2023). "P*" means the quantity is in the paper but the symbol or the argument is changed here. "O" means it is introduced in this note.

symbol	definition	origin	units
Heads and geometry			
N	number of heads in the simulated cluster. (The paper's $N(\mathrm{d}t, \mathrm{d}u)$ is a Poisson measure, unrelated.)	O	–
i	head index, $i = 1, \ldots, N$	O	–

symbol	definition	origin	units
$\alpha_i(t)$	attachment state of head i: 1 attached, 0 detached	P	–
$X_i(t)$	position of the tip of head i relative to its anchor on the myosin filament	P	nm
$h_i(t)$	signed distance from the tip to the nearest actin site, taken in $[-d/2, d/2)$; the site is at $X_i + h_i$. The paper writes $h_t^\star$ for the representative in $[-d/2, d/2)$ of $h_t \in \mathbb{T}_d$	P	nm
d	distance between two consecutive actin sites (40 nm)	P	nm
x	tip coordinate of a *detached* head, as in $K_{0\to1}(x, h)$	P	nm
x_{att}	tip coordinate of an *attached* head. It equals the position of the actin site the head is bound to, i.e. X_i when $\alpha_i = 1$. Used as the argument of the rates $K_{1\to0}$, $K_{0\to1}^{\text{rev}}$ and of w_1 (the paper writes X_{t^-} in (3.2))	O	nm
x_{det}	$x_{\text{det}} := x_{\text{att}} - h$. The detached tip position paired with x_{att} through the value h: a detached head at x_{det} with distance h to its site attaches and lands at $x_{\text{det}} + h = x_{\text{att}}$; the reverse jump from x_{att} lands at x_{det} with distance h	O	nm
Energies and thermal parameters			
$w_0(x)$, $w_1(x)$	energy of a detached tip at x, and of an attached tip at $x_{\text{att}} = x$ (Appendix A)	P	zJ
w_1', w_1''	first and second derivative of w_1	O	pN, pN/nm
k_BT	thermal energy, $k_B = 1.380649 \times 10^{-2}$ zJ/K, $T = 298$ K, so $k_BT \simeq 4.11$ zJ. (Here T is the temperature; durations are written $t_{\max}$)	P	zJ
μ_T	chemical potential of ATP, 100 zJ	P	zJ
η	micro drag of a detached tip, 0.097 pN ms/nm: $\mathrm{d}X = -\eta^{-1}w_0'(X)\mathrm{d}t + \sqrt{2\eta^{-1}k_BT}\,\mathrm{d}B$	P	pN ms/nm
Jump rates			
$K_{0\to1}(x, h)$	rate of direct attachment for a detached head at x with site distance h (channel A). Primitive, Table D.1	P	ms^{-1}
$K_{1\to0}^{\text{rev}}(x, h)$	rate of attachment by the reverse of detachment (channel B). Primitive, Table D.1	P	ms^{-1}
$K_{1\to0}(x_{\text{att}}, h)$	density in h of direct detachment of a head attached at x_{att}, landing with distance h (channel C). Derived from detailed balance	P	$\text{ms}^{-1}\text{nm}^{-1}$
$K_{0\to1}^{\text{rev}}(x_{\text{att}}, h)$	density in h of the reverse of attachment (channel D). Derived from detailed balance	P	$\text{ms}^{-1}\text{nm}^{-1}$
$k_{1\to0}(x_{\text{att}})$, $k_{0\to1}^{\text{rev}}(x_{\text{att}})$	global rates of channels C and D, $\int K\,\mathrm{d}h$ over $[-d/2, d/2)$. The paper writes $k_{1\to0}(x, 0)$ in Appendix A	P*	ms^{-1}
A, B, C, D	the four jump channels (table in Section 3); $n_A, \ldots, n_D$ are the numbers of events through each	O	–
c, e	clock and threshold of a channel; a jump occurs when $c > e$, $e \sim \mathcal{E}(1)$ (Algorithm 1)	P	–
Sliding variables			
$x_c(t)$	actin coordinate: all sites move rigidly with it, so an attached tip has $\dot{X}_i = \dot{x}_c$	P*	nm
$\dot{x}_c$	sliding velocity, < 0 in contraction	P	nm/ms
v	value imposed on $\dot{x}_c$ in the velocity clamp	O	nm/ms
u	shortening speed $u := -\dot{x}_c$; $u > 0$ shortening, $u < 0$ lengthening	O	nm/ms
$u_5(t)$	speed averaged over 5 ms, $u_5(t + 2.5\,\text{ms}) = -[x_c(t + 5\,\text{ms}) - x_c(t)]/5\,\text{ms}$	O	nm/ms
δt, Δx_c	time step (0.002 ms) and the displacement of x_c in one step	O	ms, nm

symbol	definition	origin	units
Forces			
f_i	force transmitted to the actin by head i: $f_i = \alpha_i w_1'(X_i)$ ($= 0$ if detached). $f_i > 0$ is traction. The paper's integrand is $\partial_x w_1$	P*	pN
F_{mot}	total motor force, $F_{\mathrm{mot}} = \sum_{i=1}^{N} f_i$	O	pN
$\hat{\tau}(t)$	instantaneous mean force per head over all N heads (attached or not), $\hat{\tau} = F_{\mathrm{mot}}/N$. Estimator of τ_c	O	pN
$\tau_c(t)$	the paper's force per head, $\int \partial_x w_1 \, p(t,1,x)\,\mathrm{d}x$. It is the expectation of one f_i, so $\mathbb{E}[\hat{\tau}] = \tau_c$	P	pN
$n_a(t)$	attached fraction, $N^{-1}\sum_i \alpha_i$. The paper's $\check{n}_a$ is its long-time value at $\dot{x}_c = 0$	P*	–
$\check{\tau}_c$	force per *attached* head, τ_c/n_a	P	pN
$\tau_{ss}(u)$	**steady-state mean force per head at imposed shortening speed** u: the long-time average of $\hat{\tau}$ in the velocity clamp $\dot{x}_c = -u$, equivalently τ_c at $\dot{x}_c = -u$ in steady state. Measured as the average over the N heads and over the time after a burn-in. Fig. 5.2 plots $\tau_{ss}(u)/\tau_{c,0}$ against u for $u > 0$	O	pN
$\tau_{c,0}$	isometric force per head, used to normalise. In this work $\tau_{c,0} = \frac{1}{2}[\tau_{ss}(0.1) + \tau_{ss}(-0.1)] = 0.899$ pN (the exact $\dot{x}_c = 0$ ensemble is quenched in h, see Section 9)	P*	pN
σ_f	standard deviation of one f_i	O	pN
Closure, load and device			
F_{ext}	external force on the actin, positive when it assists contraction (i.e. the force along $+x$ is $-F_{\mathrm{ext}}$)	O	pN
P	load opposing contraction, $P = -F_{\mathrm{ext}}$	O	pN
$\bar{p}$	load per head, $\bar{p} = P/N$	O	pN
ν	macroscopic drag on x_c: drag force $= -\nu\dot{x}_c$	O	pN ms/nm
$\tilde{\nu}$	drag per head, $\tilde{\nu} = \nu/N$	O	pN ms/nm
$\bar{p}_{\mathrm{ss}}(u)$	steady-state load per head that sustains speed u, $\bar{p}_{\mathrm{ss}}(u) := \tau_{ss}(u) - \tilde{\nu}u$ (eq. (8))	O	pN
K_+	total attached stiffness used in the numerical scheme, $K_+ = \sum_i \alpha_i \max(w_1''(X_i), 0)$	O	pN/nm
$\tilde{K}$	mean elastic stiffness per head, $\tilde{K} = \mathbb{E}[N^{-1}\sum_i \alpha_i w_1''(X_i)]$ (heuristic of Section 8)	O	pN/nm
θ	single relaxation time of the force $\hat{\tau}$ toward τ_{ss} (heuristic only)	O	ms
$\tau_{ss}'(u)$	derivative $\mathrm{d}\tau_{ss}/\mathrm{d}u$; $\bar{p}_{\mathrm{ss}}'(u) = \tau_{ss}'(u) - \tilde{\nu}$	O	pN ms/nm
Mixed device (spring in series)			
$x_L(t)$	far end of the spring, $x_L = x_{L,0} + Vt$	O	nm
V	speed of the far end; $V > 0$ stretches (x_L increases)	O	nm/ms
k_L	spring stiffness *per head* (the spring is Nk_L in total)	O	pN/nm
g	gap $g = x_L - x_c$	O	nm
$\bar{p}_0$	$\bar{p}_0 := k_L x_{L,0}$, the load per head at $t = 0$ for $x_c(0) = 0$; then $\bar{p}(t) = k_L g = \bar{p}_0 + k_L V t - k_L x_c$	O	pN
Run labels			
$t_{\max}$	duration of a run	O	ms
yield load	first time on the upward ramp at which $u_5 < -3\,\mu\mathrm{m\,s^{-1}}$; the load at that time, in units of $\tau_{c,0}$	O	–
recapture load	first time on the downward ramp, after $u_5 < -10\,\mu\mathrm{m\,s^{-1}}$ has occurred, at which $u_5 > -3\,\mu\mathrm{m\,s^{-1}}$; the load at that time	O	–

3 The h-model as implemented

3.1 State and energies

Head i carries (α_i, X_i, h_i), with $h_i = 0$ when attached. The energies w_0, w_1 are the regularised piecewise quadratic potentials of Table D.1 (Appendix A). A head transmits force to the actin only when attached:

$$f_i = \alpha_i \, w_1'(X_i), \qquad F_{\mathrm{mot}} = \sum_{i=1}^{N} f_i = N\hat{\tau}. \tag{1}$$

3.2 Jumps and detailed balance

Four channels, each with its own clock c and threshold $e \sim \mathcal{E}(1)$ (Algorithm 1 of the paper):

channel	transition	rate
A	$0 \to 1$ (direct attachment)	$K_{0\to1}(x, h)$
B	$0 \to 1$ (reverse of detachment)	$K^{\mathrm{rev}}_{1\to0}(x, h)$
C	$1 \to 0$ (direct detachment)	$k_{1\to0}(x_{\mathrm{att}}) = \int K_{1\to0}(x_{\mathrm{att}}, h)\,\mathrm{d}h$
D	$1 \to 0$ (reverse of attachment)	$k^{\mathrm{rev}}_{0\to1}(x_{\mathrm{att}}) = \int K^{\mathrm{rev}}_{0\to1}(x_{\mathrm{att}}, h)\,\mathrm{d}h$

A and B are active for a detached head, with $x = X_i$ and $h = h_i$. C and D are active for an attached head, with $x_{\mathrm{att}} = X_i$. $K_{0\to1}$ and $K^{\mathrm{rev}}_{1\to0}$ are the primitive rates of Table D.1. The other two follow from detailed balance. With the pairing $x_{\mathrm{det}} = x_{\mathrm{att}} - h$,

$$K^{\mathrm{rev}}_{0\to1}(x_{\mathrm{att}}, h) = \frac{1}{d}\, K_{0\to1}(x_{\mathrm{det}}, h)\, \exp\Big[\frac{w_1(x_{\mathrm{att}}) - w_0(x_{\mathrm{det}})}{k_B T}\Big], \tag{2}$$

$$K_{1\to0}(x_{\mathrm{att}}, h) = \frac{1}{d}\, K^{\mathrm{rev}}_{1\to0}(x_{\mathrm{det}}, h)\, \exp\Big[\frac{w_1(x_{\mathrm{att}}) - w_0(x_{\mathrm{det}}) + \mu_T}{k_B T}\Big]. \tag{3}$$

The factor $1/d$ is discussed in Section 9. On detachment through C or D, the landing value h' is drawn from the density $K_{1\to0}(x_{\mathrm{att}}, \cdot)$ or $K^{\mathrm{rev}}_{0\to1}(x_{\mathrm{att}}, \cdot)$ using tabulated cumulative distributions.

3.3 Time stepping

At each step $\delta t = 0.002$ ms: (i) compute F_{mot} and fix Δx_c (Section 4); (ii) update the clocks of the active channels and select the channel with the largest positive residual $c - e$, if any; (iii) move detached tips by Euler–Maruyama, $\Delta X = -\eta^{-1} w_0'(X)\delta t + \sqrt{2k_B T \delta t/\eta}\,\xi$, $\xi \sim \mathcal{N}(0,1)$, with $h \leftarrow h + \Delta x_c - \Delta X \pmod d$, and move attached tips by Δx_c; (iv) apply jumps: attachment sets $X \leftarrow X + h + \Delta x_c$, $h \leftarrow 0$; detachment sets $X \leftarrow X + \Delta x_c - h'$, $h \leftarrow h'$; (v) reset the clocks of jumping heads. Numerically, $K_{1\to0}$ and $K^{\mathrm{rev}}_{0\to1}$ are tabulated on a grid of $x_{\mathrm{att}} \in [-100, 100]$ nm (step 0.2) and h (midpoints, step 0.25 nm); exponents are clipped to ± 30 and tips to ± 100 nm.

4 Imposing the sliding

4.1 Velocity clamp (the paper's protocol)

$\dot{x}_c = v$ is prescribed. Heads are then independent, and an ensemble of N heads with random initial h_i is a Monte Carlo sample of one head. We measure $\langle\hat{\tau}\rangle$ and $\langle n_a\rangle$, averaged over the N heads and over time after a burn-in, and define

$$\tau_{ss}(u) := \langle\hat{\tau}\rangle\big|_{\dot{x}_c = -u}. \tag{4}$$

4.2 Soft device

The actin is rigid with coordinate x_c, subject to the drag $-\nu\dot{x}_c$, the external force and the motor force. All x_c-dependence of the energy $E = \sum_i \alpha_i w_1(X_i)$ comes from the attached heads ($\partial X_i/\partial x_c = 1$), so $\partial E/\partial x_c = F_{\mathrm{mot}}$. Overdamped force balance gives

$$F_{\mathrm{ext}} + F_{\mathrm{mot}} + \nu\,\dot{x}_c = 0, \qquad \text{i.e.} \qquad \nu\,\dot{x}_c = -F_{\mathrm{ext}} - \sum_{i=1}^{N} f_i. \tag{5}$$

Every term is a force in the direction of contraction: $F_{\mathrm{ext}} > 0$ assists contraction, $F_{\mathrm{mot}} > 0$ is traction, and the drag on a filament moving at shortening speed u is $-\nu u = +\nu\dot{x}_c$. Equation (5) is the same update as the δs step of the earlier N-head JAP codes, with $v \leftrightarrow \nu$ and $F \leftrightarrow -F_{\mathrm{ext}}$. If the opposite sign convention is used (force $F_{\mathrm{mot}}^{(+x)} = -\sum_i f_i$ along $+x$), the balance reads $F_{\mathrm{ext}}^{(+x)} + F_{\mathrm{mot}}^{(+x)} - \nu\dot{x}_c = 0$ and is the same equation.

Per-head form. With $\tilde{\nu} = \nu/N$, $F_{\mathrm{ext}} = -N\bar{p}$ and $F_{\mathrm{mot}} = N\hat{\tau}$,

$$\hat{\tau} = \bar{p} + \tilde{\nu}\,u. \tag{6}$$

The force on the filament is $N\hat{\tau}$, not $\hat{\tau}$.

Numerics. At fixed α, $\frac{\mathrm{d}}{\mathrm{d}t}F_{\mathrm{mot}} = K_+\dot{x}_c$ (for the clipped stiffness), which makes (5) stiff with rate K_+/ν. We use the semi-implicit step

$$\Delta x_c = -\frac{(F_{\mathrm{ext}} + F_{\mathrm{mot}})\,\delta t}{\nu + K_+\,\delta t}. \tag{7}$$

There is no thermal noise on x_c.

4.3 Mixed device

A spring of stiffness Nk_L links the actin to a far end at $x_L(t) = x_{L,0} + Vt$, giving the load per head $\bar{p} = k_L(x_L - x_c) = \bar{p}_0 + k_L V t - k_L x_c$. This is a spring in series with the cluster, like the elastic backbone k_b of Fig. 7(b) of Caruel, Allain and Truskinovsky (J. Mech. Phys. Solids 76, 2015). In that figure the force T is applied at the two ends, whereas here the far end is held or driven in position. The closure (6) is unchanged, with $F_{\mathrm{ext}} = -N\bar{p}(t, x_c)$ now depending on x_c. $V = 0$ is a fixed far end.

4.4 What is imposed and what is measured

experiment	imposed	measured
isometric	$\dot{x}_c = 0$	$\langle n_a\rangle$, $\langle\hat{\tau}\rangle$, $\check{\tau}_c$
velocity-clamp sweep	$\dot{x}_c = v$, 21 values from +6 to $-15\,\mu\mathrm{m\,s^{-1}}$	$\langle\hat{\tau}\rangle$, $\langle n_a\rangle \Rightarrow \tau_{ss}(u)$
soft-device ramp	$F_{\mathrm{ext}}(t) = -N\bar{p}(t)$, $\nu = N\tilde{\nu}$	$x_c(t)$, hence $u_5(t)$; also $\hat{\tau}$, n_a
mixed device	$F_{\mathrm{ext}} = -N[\bar{p}_0 + k_L V t - k_L x_c]$, $\nu = N\tilde{\nu}$	$x_c(t)$, and $\bar{p}(t)$ computed from it

Common settings. Velocity clamp: $N = 400$, $t_{\max} = 60$ ms, burn-in 15 ms, $\delta t = 0.002$ ms. Soft ramp: $\bar{p}(t)$ is piecewise linear, held at 0.22 pN for 60 ms, raised to 1.144 pN over 300 ms and lowered back over 300 ms ($0.245 \to 1.272$ in units of $\tau_{c,0} = 0.899$ pN); $\tilde{\nu} \in \{0.05, 0.30\}$ pN ms/nm and $N \in \{30, 100, 300\}$. All runs start with every head detached, h uniform on $[-d/2, d/2)$ and the tip drawn from a Gaussian around $x = -2$ nm.

5 Steady states under force control and the S

5.1 Steady-state relation

In a steady state at speed u the mean force per head is $\tau_{ss}(u)$, so (6) gives the load that sustains that speed:

$$\bar{p}_{\rm ss}(u) = \tau_{ss}(u) - \tilde{\nu}\, u. \tag{8}$$

For $\tilde{\nu} \to 0$ this is the paper's curve, which corresponds to the experimental force-clamp protocol. A positive $\tilde{\nu}$ shifts each point horizontally by $-\tilde{\nu}u$.

5.2 Stability (heuristic)

Assume the head population relaxes to τ_{ss} with a single time θ, $\theta\dot{\hat{\tau}} = \tau_{ss}(u) - \hat{\tau}$, with $u = (\hat{\tau} - \bar{p})/\tilde{\nu}$ from (6). Linearising,

$$\theta\, \delta\dot{\hat{\tau}} = \Big(\frac{\tau'_{ss}(u)}{\tilde{\nu}} - 1\Big)\delta\hat{\tau}, \tag{9}$$

so a steady state is stable if and only if $\tau'_{ss}(u) < \tilde{\nu}$, equivalently $\bar{p}'_{\rm ss}(u) < 0$. This is a one-timescale argument, a criterion for the sign and not for the location.

5.3 Origin of the fold

Where τ_{ss} is non-monotone, $\bar{p}_{\rm ss}(u)$ is multivalued as a function of the load. We measured τ_{ss} decreasing monotonically for lengthening ($u < 0$), and a slight rise for small $u > 0$ (from 0.88 to 0.94 pN, comparable to the scatter), as the paper also notes for small shortening speeds. For $\tilde{\nu} = 0.05$ this leaves a stable shortening branch ending at $\bar{p}_{\rm ss} \simeq 1.03\,\tau_{c,0}$ ($u \simeq 0.2$), an unstable stretch around the stall point, and a fast-lengthening branch $u \simeq (\hat{\tau} - \bar{p})/\tilde{\nu}$ whose minimum load is $\simeq 0.59\,\tau_{c,0}$ at $|u| \simeq 3$. Increasing $\tilde{\nu}$ removes the fold (Table 2).

$\tilde{\nu}$ (pN ms/nm)	max of $\bar{p}_{\rm ss}$ on stall/shortening side	min of $\bar{p}_{\rm ss}$ on lengthening side
0.05	$1.03\,\tau_{c,0}$ at $u = 0.2$	$0.59\,\tau_{c,0}$ at $u = -3$
0.15	$1.02\,\tau_{c,0}$	$0.85\,\tau_{c,0}$ at $u = -1.5$
0.30	$1.07\,\tau_{c,0}$	$0.98\,\tau_{c,0}$ at $u = -0.7$

Table 2: Fold positions obtained from (8) with the measured τ_{ss} (linear interpolation, $\tau_{c,0} = 0.899$ pN). For $\tilde{\nu} = 0.30$ the fold depth, $0.09\,\tau_{c,0}$, is of the order of the scatter of τ_{ss}.

6 What the figure shows

Left panel (simulation, velocity clamp). Horizontal axis: $\tau_{ss}(u)/\tau_{c,0}$. Vertical axis: u in μm/s; $u > 0$ is shortening and $u < 0$ is lengthening. Each marker is one clamp run of $N = 400$ heads with $\dot{x}_c = -u$ imposed. The point at $u = 0$ is replaced by the mean of $u = \pm 0.1$ (definition of $\tau_{c,0}$).

Middle panel (no new simulation). It is the transformation (8) applied to the left-panel data, with the horizontal axis $\bar{p}_{\rm ss}(u)/\tau_{c,0}$. Each line is one $\tilde{\nu}$; the black line is $\tilde{\nu} = 0$ and equals the left panel. The horizontal axis is cut to $0 \le \bar{p}_{\rm ss}/\tau_{c,0} \le 1.5$.

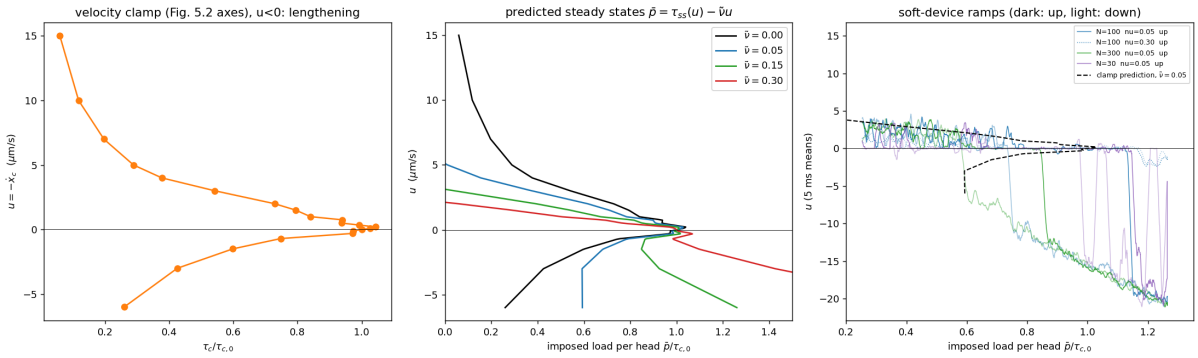


Figure 1: Left: velocity-clamp curve. Middle: steady-state load (8). Right: soft-device ramps.

Right panel (simulation, soft device). Horizontal axis: the imposed load $\bar{p}(t)/\tau_{c,0}$ at the midpoint of a window. Vertical axis: u_5. Dark lines are the upward ramp, light lines the downward ramp; solid is $\tilde{\nu} = 0.05$ (colours: $N = 30, 100, 300$), dotted is $\tilde{\nu} = 0.30$ with $N = 100$. The dashed black line is the middle-panel prediction for $\tilde{\nu} = 0.05$. Here the load is the input and u_5 is an output, so jumps appear as vertical segments.

7 Results

7.1 Validation

- *Thermodynamic consistency.* At $\mu_T = 0$ with $\dot{x}_c = 0$ the net cycling vanishes: $(n_A - n_D)/(n_A + n_D) = 0.002$ and $(n_C - n_B)/(n_C + n_B) = -0.003$ ($N = 600$, 90 ms). At $\mu_T = 100$ zJ the net flux is positive.
- *Isometric indicators* (Table 5.1: $\check{n}_a = 0.151$, $\check{\tau}_c = 6.39$ pN). We obtain $n_a \simeq 0.14$–0.15 and $\check{\tau}_c \simeq 5$–6 pN, depending on the seed. The spread reflects the quenched h distribution.
- *Force–velocity curve (Fig. 5.2).* $\tau_{ss}/\tau_{c,0} \simeq 0.29$ at $u = 5\,\mu\mathrm{m\,s^{-1}}$ and $\simeq 0.06$ at $15\,\mu\mathrm{m\,s^{-1}}$, which agrees with the curve as read from the figure.
- *Closure.* In the soft-device runs $\bar{p} + \tilde{\nu}u = \hat{\tau}$ holds to within 0.003 pN when each term is averaged over a 40 ms window.

7.2 Velocity-clamp data

7.3 Soft-device ramps

The upward ramp follows the shortening branch, stalls ($u \approx 0$) close to $\tau_{c,0}$, and then yields to lengthening at speeds up to $\simeq 21\,\mu\mathrm{m\,s^{-1}}$ ($\simeq (\hat{\tau} - \bar{p})/\tilde{\nu}$, with few heads attached). On the way down the filament stays on the fast branch until a load that depends on N. For $\tilde{\nu} = 0.30$ there is no runaway.

7.4 Dependence on N

Equation (8) is the $N \to \infty$ limit. At finite N, $u = (\hat{\tau} - \bar{p})/\tilde{\nu}$ inherits the fluctuations of $\hat{\tau}$, of order $\sigma_f/\sqrt{N}$ (a Bernoulli estimate gives $\sigma_f \approx \sqrt{n_a(1-n_a)}\,\check{\tau}_c \approx 2$ pN). The fluctuation of the speed is then $\delta u \sim \sigma_f/(\tilde{\nu}\sqrt{N})$, about $4\,\mu\mathrm{m\,s^{-1}}$ for $N = 100$ and $\tilde{\nu} = 0.05$ (an order-of-magnitude estimate), comparable to the mean speed. This is consistent with what we observe: at $N = 100$, $\tilde{\nu} = 0.05$ and $\bar{p} = 0.28$ pN the mean speed is $2.45\,\mu\mathrm{m\,s^{-1}}$ with $\hat{\tau} = 0.41$ pN and $n_a = 0.14$,

u	$\langle\hat{\tau}\rangle$ (pN)	$\langle n_a\rangle$	u	$\langle\hat{\tau}\rangle$ (pN)	$\langle n_a\rangle$
-6	0.233	0.034	0.35	0.892	0.145
-3	0.382	0.056	0.5	0.843	0.134
-1.5	0.538	0.082	0.75	0.845	0.139
-0.7	0.672	0.106	1	0.756	0.129
-0.3	0.874	0.140	1.5	0.715	0.138
-0.1	0.875	0.142	2	0.655	0.140
0 (quenched)	0.661	0.121	3	0.488	0.133
0.1	0.923	0.148	4	0.339	0.129
0.2	0.938	0.154	5	0.260	0.117
			7	0.177	0.092
			10	0.106	0.079
			15	0.053	0.059

Table 3: Velocity clamp, $N = 400$, one run per speed; u in $\mu\mathrm{m\,s^{-1}}$. The table lists $\langle\hat{\tau}\rangle$, the measured value of $\tau_{ss}(u)$. Errors were not estimated by repetition; the scatter between neighbouring speeds (for example $u = -0.3$ and -0.7) suggests ± 0.05–0.1 pN.

N	$\tilde{\nu}$	yield load	recapture load	remark
30	0.05	$1.15\,\tau_{c,0}$	intermittent	rescues up to $1.23\,\tau_{c,0}$, no clean loop
100	0.05	$1.14\,\tau_{c,0}$	$0.74\,\tau_{c,0}$	hysteresis loop
300	0.05	$0.85\,\tau_{c,0}$	$0.59\,\tau_{c,0}$	recapture at predicted floor
100	0.30	none	none	max lengthening speed $2.4\,\mu\mathrm{m\,s^{-1}}$, reversible

Table 4: Yield and recapture loads as defined in Section 2, in units of $\tau_{c,0} = 0.899$ pN. One realisation per row. Predicted for $\tilde{\nu} = 0.05$, $N \to \infty$: yield $1.03\,\tau_{c,0}$, floor $0.59\,\tau_{c,0}$.

whereas the clamp curve gives $\tau_{ss}(2.45) \simeq 0.58$ pN at the same n_a. The mean force per attached head is lower in the soft device because heads experience a fluctuating speed.

Two trends are supported by the data: the recapture load moves toward the predicted floor as N grows ($0.74 \to 0.59$), and the $N = 30$ system does not stay on either branch. The yield loads $(1.15, 1.14, 0.85)$ are not monotone in N and come from single runs, so no trend is claimed for them.

8 Search for oscillations

8.1 Fixed far end and constant load

Runs of 300 ms, $\tilde{\nu} = 0.05$ pN ms/nm unless stated, one realisation each. $\bar{p}$ is the load per head; a fixed far end ($V = 0$) gives $\bar{p} = \bar{p}_0 - k_L x_c$.

N, load	result	reading
30, constant 0.72 pN	7 lengthening bursts, mean interval 37 ms, s.d. 36 ms	irregular stick–slip (noise-driven)
100, $\bar{p}_0 = 0.90$, $k_L = 0.04, 0.10$	x_c stays within a few nm of stall	pinned at stall
300, $\bar{p}_0 = 0.90$, $k_L = 0.04$	step-like slips of a few nm, autocorrelation maximum 0.15 at 18 ms	no periodicity
30, $\bar{p}_0 = 0.90$, $k_L = 0.02, 0.04, 0.10$	1–3 bursts in 240 ms	rare slips
100, $\tilde{\nu} = 0.02$, $\bar{p}_0 = 0.95$, $k_L = 0.01$	5 bursts, interval 54 ms, s.d. 22 ms, x_c drifts by +45 nm	more regular than Poisson; not a demonstrated limit cycle

With a fixed far end no sustained oscillation was obtained. A two-variable heuristic for a stall point with elastic load k_L (my own, untested against the simulation) uses θ, $\tilde{K}$ and $\tau'_{ss}(0)$ as defined in Section 2. It gives $\det = k_L/(\tilde{\nu}\theta) > 0$ and a Hopf instability when

$$\tau'_{ss}(0) > \theta\,(\tilde{K} + k_L) + \tilde{\nu}. \tag{10}$$

For a constant load ($k_L = 0$) $\det = 0$ and the heuristic excludes a Hopf bifurcation. A length-step test to measure θ and $\tilde{K}$ was inconclusive (the force rose after a release, which I traced to the structure of w_1 in Section 9), so the condition has not been evaluated.

8.2 Driven far end: stick–slip in the mixed device

Now $V > 0$: the far end is stretched at speed V, and $\bar{p}(t) = \bar{p}_0 + k_L V t - k_L x_c$.

Mechanism. In steady lengthening at speed V ($u = -V$), (8) gives the sustaining load $\bar{p}_{\rm ss}(-V) = \tau_{ss}(-V) + \tilde{\nu}V$. Our measured τ_{ss} decreases with lengthening speed (Table 3), so for $\tilde{\nu} = 0.05$ this load decreases with V for roughly $0.3 \lesssim V \lesssim 6\,\mu{\rm m\,s^{-1}}$. Suppose the speed follows the branch $\bar{p}_{\rm ss}(u) = k_L g$ quasi-statically, with $g = x_L - x_c$. Then $\dot{g} = V + u$ and $\delta u = (k_L/\bar{p}'_{\rm ss}(u))\,\delta g$, so

$$\delta\dot{g} = \frac{k_L}{\bar{p}'_{\rm ss}(u)}\,\delta g, \tag{11}$$

which grows when $\bar{p}'_{\rm ss}(u) > 0$. This is the same unstable condition as in Section 5, and it holds for any $k_L > 0$. The stable parts are the stalled state and the fast-lengthening branch, so the expected motion is stick–slip: the load rises while the heads hold ($u \approx 0$), the filament slips, and the load falls until the heads recapture. The argument ignores the head kinetics.

Runs. $\tilde{\nu} = 0.05$, $\bar{p}_0 = 0.50$ pN, $t_{\max} = 200$ ms, one realisation each. The load is computed from $x_c(t)$. A "peak" is a maximum of the 1 ms-smoothed $\bar{p}$ above $1.0\,\tau_{c,0}$ separated from the previous peak by more than 5 ms; the FFT is of $\bar{p}$ after 40 ms.

N	k_L	V	load peaks	interval (ms)	mean peak, FFT period
100	0.04	2.0	10	16.4 ± 6.0	$1.35\,\tau_{c,0}$, 40 ms
100	0.04	0.7	9	17.5 ± 13.3	$1.26\,\tau_{c,0}$, 53 ms
100	0.10	2.0	15	10.8 ± 2.9	$1.21\,\tau_{c,0}$, 10 ms
300	0.04	2.0	6	21.5 ± 9.8	$1.15\,\tau_{c,0}$, 12 ms

In all four runs the load repeatedly sweeps from about $0.3\,\tau_{c,0}$ to above $1.2\,\tau_{c,0}$, and the filament slips at up to about $-20\,\mu\mathrm{m\,s^{-1}}$ each time. The cycles are noisy: the coefficient of variation of the peak intervals is 0.27 to 0.76, and the FFT period and the mean interval do not agree in every run. The strongest case is $k_L = 0.10$, $V = 2$ (CV 0.27, FFT period 10 ms, interval 10.8 ms). The number of slips with $u_{0.25} < -6\,\mu\mathrm{m\,s^{-1}}$ ($u_{0.25}$ being the speed averaged over 0.25 ms) also increases with V (22 for $V = 2$, 10 for $V = 0.7$, $N = 100$, $k_L = 0.04$), as expected for stick–slip. What is established here is repeated, load-synchronised slipping; a clean limit cycle (narrow spectral line) is not.

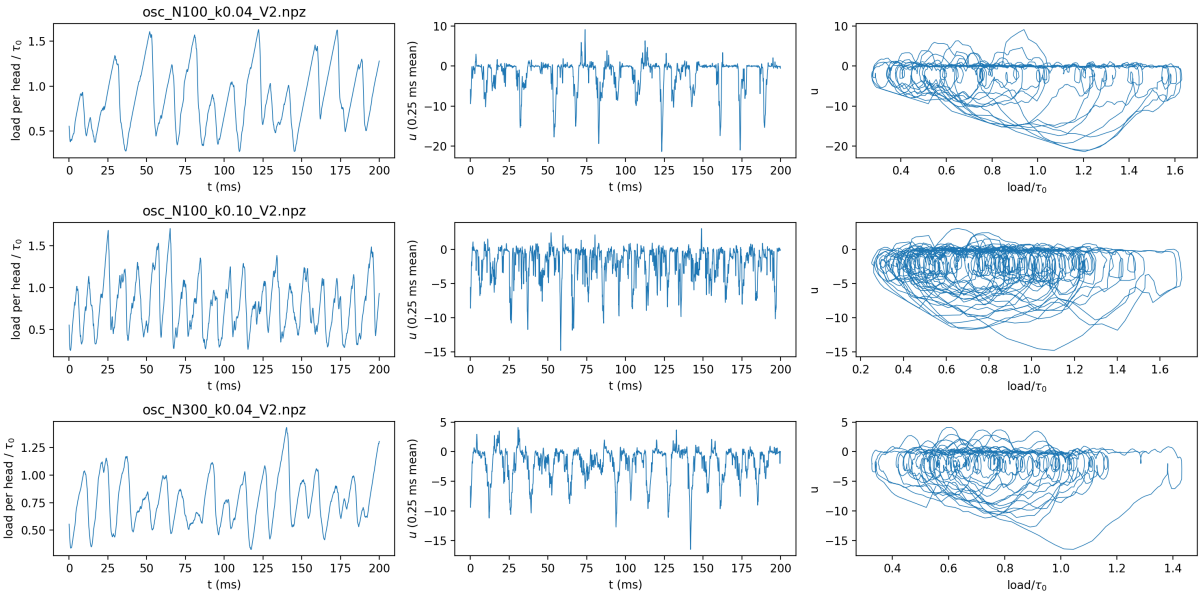


Figure 2: Driven far end, $V = 2\,\mu\mathrm{m\,s^{-1}}$. Top: $N = 100$, $k_L = 0.04$. Middle: $N = 100$, $k_L = 0.10$. Bottom: $N = 300$, $k_L = 0.04$. Left: load per head $\bar{p}/\tau_{c,0}$ against time. Middle: speed (0.25 ms means). Right: speed against load.

9 Caveats and limitations

1. **Factor $1/d$ in the detailed-balance rates.** Taken literally, (3.4)–(3.5) of the paper give $n_a = 0.03$. Since $K_{0\to1}$ and $K^{\mathrm{rev}}_{1\to0}$ are rates while $K_{1\to0}$ and $K^{\mathrm{rev}}_{0\to1}$ are densities in h, the relation needs a length, and for the uniform measure on the torus it is d (parameter `H_NORM`). With it, n_a matches Table 5.1. The $\mu_T = 0$ test passes for any value of this factor, so it does not validate it.

2. **Potentials.** Evaluated from Table D.1 as extracted from the PDF text, w_1 and w_0 agree with Fig. 5.1(a) in the difference $w_0 - w_1$ at the minima (≈ 77 zJ) but differ by a constant of about 30 zJ in absolute value. A constant cancels in all rates. I did not compare the curves point by point. *In addition*, inside the regularisation window $[x_\ell, x_r] = [1.3, 2.5]$ nm the transcribed interpolation (Hermite terms plus the bump $\breve{w}\,\varphi_5$ with $\breve{w} = 5.7$ zJ) gives w_1' between -9 and $+23$ pN and w_1'' between -71 and $+49$ pN/nm. Attached heads sit at $x_{\mathrm{att}} \simeq 2.2 \pm 3.2$ nm, so this window is populated. I could not check against Fig. 5.1(a) whether the paper's potential has this structure. All force-dependent results here (stall force, yield and recapture loads, the elastic response) are for the potential as transcribed.

3. **Table D.1 reading.** The box in x of $K_{0\to1}$ is taken as $[x_\ell^+, x_r^+] = [1.5, 17]$ nm (a positive interval, as in the text of Section 5.1); the exponent in $K^{\mathrm{rev}}_{1\to0}$ is clipped to ±30 for numerical

safety. The paper's text and Table D.1 are not fully consistent about which of $K_{1\to0}$, $K^{\mathrm{rev}}_{1\to0}$ is the primitive rate; I took the table at face value.

4. **Isometric normalisation.** At $\dot{x}_c = 0$ the ensemble is quenched in h (a head whose site is more than about 4 nm away rarely attaches), so $\langle\hat{\tau}\rangle$ at $v = 0$ depends on the sample (0.66 pN against 0.88–0.94 pN at $|u| = 0.1$). We therefore define $\tau_{c,0}$ from $u = \pm 0.1$.

5. **Statistics.** One run per velocity and per $(N, \tilde{\nu})$ in the ramps; no repeated runs, no confidence intervals, and no time-step convergence study (all runs use $\delta t = 0.002$ ms).

6. **Free parameters.** $\tilde{\nu}$, k_L, V and $\bar{p}_0$ are not given by the paper; the values used are my choices.

7. **No noise on x_c.** The closure is deterministic, so at finite N and $\mu_T = 0$ the stationary law is not the soft-ensemble Gibbs measure.

8. **Heuristics.** The stability criterion of Section 5, the stick–slip argument and the Hopf condition use quasi-static or single-relaxation-time assumptions, and the δu estimate is of order of magnitude only.

9. **Range.** The clamp sweep reaches $u = -6\,\mu\mathrm{m\,s^{-1}}$ on the lengthening side, whereas the ramps explore lengthening up to $-21\,\mu\mathrm{m\,s^{-1}}$.

10 Reproducing

file	role
`hmodel_soft.py`	engine: potentials, rates, tables, `run(...)` with modes `clamp` and `soft`
`sweep_clamp.py`	velocity-clamp sweep, writes `clamp_sweep.npy`
`soft_ramp.py N nu`	force ramp up and down, writes `ramp_*.npz`
`plot_S.py`	builds Fig. 1 from the two outputs
`osc_run.py`	mixed device (spring, far end fixed or driven), writes `osc_*.npz`
`an_osc3.py`	slip statistics and Fig. 2 from `osc_*.npz`

In `soft_ramp.py` the reference force is hard-coded as 0.88 pN to define the ramp limits; the measured $\tau_{c,0}$ is 0.899 pN, and all normalised results here use the latter. `an_osc3.py` also uses 0.899 pN.

A Potentials and parameters

Each energy is $w_\alpha(x) = E_\alpha + \omega_\alpha(x)$ with, writing $L = x_r - x_\ell$ and $\kappa_\ell, \kappa_r, \tilde{x}_\ell, \tilde{x}_r, x_\alpha, x_\ell, x_r, \check{w}_\alpha, E_\alpha$ for the nine parameters of Table D.1,

$$\omega_\alpha(x) = \begin{cases} \frac{\kappa_\ell}{2}(x - \tilde{x}_\ell)^2 + \tilde{w}_\alpha, & x < x_\ell, \\ \frac{\kappa_r}{2}(x - \tilde{x}_r)^2, & x > x_r, \\ \omega^{\mathrm{int}}_\alpha(x), & x_\ell \le x \le x_r, \end{cases} \tag{12}$$

$$\begin{aligned} \omega^{\mathrm{int}}_\alpha(x) = {} & \left[\tfrac{\kappa_\ell}{2}(x_\ell - \tilde{x}_\ell)^2 + \tilde{w}_\alpha\right]\varphi_1 + \kappa_\ell(x_\ell - \tilde{x}_\ell)\varphi_2 \\ & + \tfrac{\kappa_r}{2}(x_r - \tilde{x}_r)^2\varphi_3 + \kappa_r(x_r - \tilde{x}_r)\varphi_4 + \check{w}_\alpha\varphi_5, \end{aligned} \tag{13}$$

$$\tilde{w}_\alpha = -\tfrac{\kappa_\ell}{2}(x_\alpha - \tilde{x}_\ell)^2 + \tfrac{\kappa_r}{2}(x_\alpha - \tilde{x}_r)^2, \tag{14}$$

$$\varphi_1 = \tfrac{(x-x_r)^2}{L^2}\Big[\tfrac{2(x-x_\ell)}{L}+1\Big], \qquad \varphi_2 = \tfrac{(x-x_r)^2}{L^2}(x-x_\ell), \tag{15}$$

$$\varphi_3 = \tfrac{(x-x_\ell)^2}{L^2}\Big[3-\tfrac{2(x-x_\ell)}{L}\Big], \qquad \varphi_4 = \tfrac{(x-x_\ell)^2}{L^2}(x-x_r), \tag{16}$$

$$\varphi_5 = \tfrac{\varphi_1\varphi_3}{\varphi_1(x_\alpha)\varphi_3(x_\alpha)}. \tag{17}$$

(The $\varphi_{1..4}$ are the cubic Hermite basis, so the piecewise function is C^1; φ_5 is the bump of height $\check{w}_\alpha$ at x_α that vanishes with its derivative at x_ℓ and x_r.)

parameter	w_1 ($\alpha=1$)	w_0 ($\alpha=0$)
κ_ℓ (pN/nm)	0.60	0.12
κ_r (pN/nm)	1	1
$\tilde{x}_\ell$ (nm)	-12.5	-2
$\tilde{x}_r$ (nm)	-1.5	2
x_α (nm)	2	-2
x_ℓ (nm)	1.3	-3.5
x_r (nm)	2.5	0.5
$\check{w}_\alpha$ (zJ)	5.7	5.7
E_α (zJ)	50	70

Other parameters: $d = 40$ nm, $T = 298$ K, $\mu_T = 100$ zJ, $\eta = 0.097$ pN ms/nm.

B Rate formulas

Let $b(h;\lambda,a) = \frac{1}{2}[1+\tanh\lambda(h+a)]$ for $h<0$ and $\frac{1}{2}[1-\tanh\lambda(h-a)]$ for $h\geq 0$. The primitive rates are

$$K_{0\to1}(x,h) = k_+\, b(h;\lambda_{h,+},h_+)\,\chi(x), \tag{18}$$

$$\chi(x) = \begin{cases} \frac{1}{2}[1+\tanh\lambda_{x,+}(x-x_{\ell,+})], & x < \frac{1}{2}(x_{\ell,+}+x_{r,+}), \\ \frac{1}{2}[1-\tanh\lambda_{x,+}(x-x_{r,+})], & \text{otherwise}, \end{cases} \tag{19}$$

$$\begin{aligned} K^{\mathrm{rev}}_{1\to0}(x,h) = {} & k_m\,\tfrac{1}{2}[1-\tanh\lambda_{x,m}(x-x_m)]\,b(h;\lambda_{h,m},h_m) \\ & + \Big\{k_p\,\tfrac{1}{2}[1-\tanh\lambda_{x,p}(x-x_p)]\,b(h;\lambda_{h,p},h_p) + k_b\,b(h;\lambda_{h,b},h_b)\Big\} \\ & \times e^{[w_0(x)-w_1(x+h)-\mu_T]/k_BT}. \end{aligned} \tag{20}$$

rate	parameters (rates in ms^{-1}, lengths in nm, λ in nm^{-1})
$K_{0\to1}$	$k_+ = 0.413$, $\lambda_{h,+} = 1.6$, $h_+ = 3.9$, $\lambda_{x,+} = 8$, $x_{\ell,+} = 1.5$, $x_{r,+} = 17$
$K^{\mathrm{rev}}_{1\to0}$, first term	$k_m = 0.206$, $\lambda_{h,m} = 5$, $h_m = 2$, $\lambda_{x,m} = 10$, $x_m = -28$
$K^{\mathrm{rev}}_{1\to0}$, second term	$k_p = 3.1$, $\lambda_{h,p} = 5$, $h_p = 2$, $\lambda_{x,p} = 20$, $x_p = 0.5$
$K^{\mathrm{rev}}_{1\to0}$, third term	$k_b = 0.07$, $\lambda_{h,b} = 5$, $h_b = 0.2$

The derived rates $K^{\mathrm{rev}}_{0\to1}$ and $K_{1\to0}$ are given in Section 3.